

IM3180 Design and Innovation Project (AY2025/26 Semester 1)

Individual Report

Name: Chua Jia Ying, Chloe

Group: IE08

Project Title: DeciWise

Contributions to the Project and Report

Project

Database design and setup

- Designed the schema for the MongoDB database, including the collections required
- Setup the initial database on MongoDB Atlas
- Created test admin and trainee accounts for groupmates to test

Backend Development of non AI components

- User Authentication and management:
 - Input sanitation, email formatting, and password hashing
 - Ensured role-based access control
 - Retrieved user in other pages after logging in
- Analytics report
 - End points for fetching reports according to the trainee's email
 - Saving the report to the database
 - Displaying all the reports in a table for the admins
 - Creating a schema that fits what is in the reports
- API handling
 - Added console logs to make it easier to debug
 - Cleaned up error messages and API responses to reduce confusion
 - Fixed CORS issues between react and Flask API

Backend-Frontend Integration

- Integrated backend API endpoints into frontend pages (admin dashboard, trainee page, case studies, trainer team, trainee reports, trainee case studies)
- Troubleshoot issues like: Navigation problems, Incorrect state handling
- UI Improvements
 - Updated pages to adapt to different screen sizes
 - Made the UI more consistent throughout the web application

- Access separation: trainees cant access any admin-only pages and vice versa (like case studies for admin and trainee is different)

Report (indicate pages and appendixes written by you for the Group Report)

1. Group report, pg. 14 Section 4.1.3: Tech stack for database, explaining database choice, structure and rationale
2. Group Report Appendix:
 - a. Pg. 45, Set up and installation guide
 - b. Pg. 46, Admin guide
 - c. Pg. 51, Trainee guide
 - d. Pg. 53, Administrative & maintenance guide

Reflection on Learning Outcome Attainment

Reflect on your experience during your project and the achievements you have relating to at least two of the points below:

- (a) Engineering knowledge
- (b) Problem Analysis
- (c) Investigation
- (d) Design/development of Solutions
- (e) Modern Tool Usage
- (f) The Engineer and the World
- (g) Ethics
- (h) Individual and Collaborative Team Work
- (i) Communication
- (j) Project Management and Finance
- (k) Lifelong Learning

Point 1: Modern Tool Usage

Initially, I did not know how to use Github, MongoDB, react and was unfamiliar with Flask and FastAPI. I had to learn Git properly to ensure I wouldn't break anyone's work or mess anything up, and researched React DevTools to understand the codes that my teammates had written. For the first 7-8 weeks, I actually used FastAPI to set up the backend because when I was doing research, most people recommended this. Later on, when my teammate was integrating AI into the web app, they told me it wasn't compatible with FastAPI, so I had to migrate to Flask. Thus, from this I learnt that we should always choose our tools based on the project requirements. I also got better at debugging using the browser DevTools. Overall, I became much more confident in learning new tools and now I better understand how to choose the techstack and appropriate tools according to the project.

Point 2: Individual and collaborative teamwork

Since I was not familiar with React, I found integrating the backend and frontend challenging at first. But thankfully, my teammates included comments to their codes, which helped me understand their components better. I also didn't know how to properly set up a development project from scratch, so I found it very confusing at the start. Especially when my groupmates were discussing the techstack and figuring out the technical aspects of the project, like how to store images. Since I was the only one working on the backend, I had to figure most things out myself. Our weekly check-ins eventually gave me space to ask what I should help with or if there are any backend endpoints that the front end required. Working with others also forced me to stay organised and adapt quickly to changes. Through this, I learned to communicate more proactively and also saw how important documentation and communicating can be for the team.

Acknowledging/Declaring the Use of GAI

Please refer to NTU's Current Policy & Guidelines on the Use of Generative AI available in NTUlearn home page and the link:

[https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-\(GAI\)-Tools-on-Education.aspx](https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-(GAI)-Tools-on-Education.aspx)

- Complete the following declaration if applicable.
- Create a Paper Trail to document the input prompt, output obtained, and how you have used it

I Chua Jia Ying, Chloe,
chua1094@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission:

1. Name of course: Design & Innovation Project
2. Course Code: IM3180
3. Instructor: Prof Andy Khong
4. Title of Assignment/Project Submission: DIP Individual Report

In relation to the foregoing I hereby declare that, fully and properly in accordance with the Assignment/Project Instructions I have (check where appropriate):

- i. Used GAI as permitted to assist in generating key ideas only. ☐
- ii. Used GAI as permitted to assist in generating a first text only. ☐
- And/or
- iii. Used GAI to refine syntax and grammar for correct language submission only. ☐
- Or
- iv. As it is not permitted: Not used GAI assistance in any way in the development or generation of this assignment or project. ☒

I also declare that I have :

- a. Fully and honestly submitted the digital paper trail required under the assignment/project instructions; and that
- b. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote; and that,
- c. Apart from the foregoing notices, the submission is wholly my own work.



Chloe Chua

14/11/2025

Student Name & Signature

Date